

Does Protecting the Router in Quantized MoE Improve Quality, and How?

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Abstract

Mixture-of-Experts (MoE) models spend almost all of their parameters in expert networks, and only a tiny fraction in the routers that choose them. We ask a narrow question: in calibration-free, weight-only post-training quantization, does leaving just those routers in BF16 protect routing and language-model quality? On OLMoE-1B-7B and Qwen1.5-MoE-A2.7B, at INT8, INT4 and INT3, the answer is yes for routing. Protecting the routers reduces routing KL and top-1 expert flips relative to quantizing them, with disjoint bootstrap intervals. A parameter-count-matched placebo does nothing. We then split the drift that remains between its two possible causes, the error in the activations reaching the router and the rounding of the router’s own weights, to see which contributes more: the activations dominate at INT8, the rounded weights at INT4. Router protection is cheap and specific.

1 Introduction

A Mixture-of-Experts layer sends each token through a small linear *router* and then through only a handful of *experts* (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). The experts are almost the whole model. The routers are not: 0.030% of OLMoE-1B-7B and 0.021% of Qwen1.5-MoE-A2.7B. Quantizing the experts therefore saves nearly everything. Quantizing the routers as well can send a token to the wrong experts.

We test one safeguard, with no calibration: keep every router in BF16 and quantize every expert. The comparison is a single variable. UNIFORM quantizes routers and experts at the same INT- N width; MIXED quantizes the same experts and leaves the routers untouched; GOLD is the BF16

reference. A parameter-count-matched PLACEBO then asks whether protecting any equally small set of weights would have done as well.

This is not a new proposal. Fu et al. (2025) already run similar experiments and Fang and Huang (2026) argue that freezing the router is not enough. What those papers do not isolate is the calibration-free, weight-only version of the variable, scored on routing itself, with a control and a mechanism split. That is the measurement we report.

Three findings, the first two previewed in Figure 1. First, MIXED beats UNIFORM on routing KL and top-1 expert flips in every model $\times$ bit-width cell we ran. Second, the gain is specific to the routers: the placebo lands on UNIFORM. Third, we split the drift that remains between the hidden states arriving at the router and the rounding of the router’s own weights (Section 6), and find that the hidden states are the larger source of flips at INT8, while for INT4 the router’s own weights are the main cause.

2 Background

MoE routing. At each MoE layer a linear router maps a hidden state $x \in \mathbb{R}^d$ to E logits. The layer keeps the top- k experts and mixes their outputs with a softmax over those k scores:

$$y = \sum_{i \in \text{Top}k(Wx)} \text{softmax}(Wx)_i E_i(x). \quad (1)$$

A flip of the top-1 index, or of the top- k set, changes which parameters compute y . OLMoE uses $E=64$, $k=8$; Qwen1.5-MoE uses $E=60$, $k=4$, plus a shared expert that we do *not* protect (Section 4).

Post-training quantization. We use weight-only, per-output-channel symmetric INT- N quan-

Protecting the router alone, across two MoE architectures

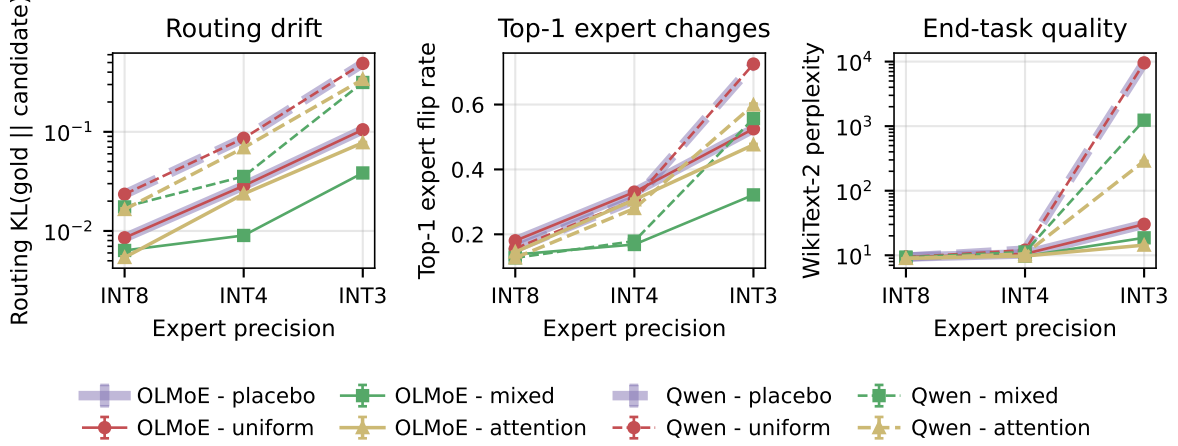


Figure 1: Routing KL (left, log scale), top-1 expert flip rate (centre) and WikiText-2 perplexity (right, log scale) against expert precision. MIXED keeps every router in BF16; UNIFORM quantizes the routers too; PLACEBO protects a parameter-count-matched non-router module instead. Error bars are 95% bootstrap intervals over sequences. PLACEBO sits on UNIFORM, that coincidence is the control result. The extra series, ATTENTION, is a follow-up that also leaves attention in BF16 (Section 5); it is not a substitute for protecting the router.

tization through TorchAO (Or et al., 2025). Each output channel w of a linear layer gets one scale s , set from the largest magnitude in that channel, and every weight in it is replaced by $s \cdot \text{round}(w_j/s)$, clipped to the 2^N signed integers. A channel can therefore hold at most 2^N distinct values: 256 at INT8, 16 at INT4, 8 at INT3. We call that count the *level ceiling* and use it in Section 4 to check what was quantized. The scale comes from the weight tensor itself: there is no calibration set, no rotation, and no reconstruction loss. Embeddings and the language-model head stay in BF16 in every policy. This is a different regime from GPTQ, AWQ, SmoothQuant or LLM.int8() (Frantar et al., 2023; Lin et al., 2024; Xiao et al., 2023; Dettmers et al., 2022), all of which fit something to data.

Metrics. For one token at one layer, let p be the softmax over all E router logits under GOLD and q the same softmax under a candidate policy, with top- k sets S_p and S_q . We report three routing quantities, each averaged over tokens and layers: the *routing KL* $\text{KL}(p \parallel q)$; the *top-1 flip rate*, the fraction of tokens whose highest-scoring expert changes; and the top- k Jaccard distance $1 - |S_p \cap S_q| / |S_p \cup S_q|$. All three are zero when routing is unchanged and grow as it drifts. WikiText-2 perplexity (Merity et al., 2017) is the end-task measure: the exponential of the mean negative log-likelihood over every scored token.

Uncertainty and the decision gate. Routing metrics are computed on a fixed subset of sequences, so we attach 95% percentile bootstrap intervals that resample whole sequences rather than tokens: tokens within a sequence are correlated, and resampling them independently would understate the uncertainty. We call the routing result positive only when the MIXED and UNIFORM intervals are disjoint on both routing KL and top-1 flip rate, and refer to that rule as the *decision gate*. Perplexity is a point estimate over every scored token; it carries no confidence interval, and the gate does not use it.

3 Related Work

Router precision in MoE quantization. Protecting the router is already standard practice. EAQuant runs experts at W4A4 (4-bit integer weights and activations) with the router at W8A8 (Fu et al., 2025). On the same OLMoE checkpoint we use, their Table 4 holds experts at W3A4 and moves the router from W3A4 to W8A8; WikiText-2 perplexity falls by 0.32 and average zero-shot accuracy rises by about one point. That is a real router-precision effect, and we do not claim otherwise. It is also confounded in three ways this paper removes. Every condition moves router *weight and activation* precision together; the baseline is a calibrated rotation method; and the most-protected router is W8A8, never a router exempt from quan-

tization. The table is scored only on end-task metrics, so it cannot say whether routing itself changed. GEMQ independently notes that routers are under 0.04% of MoE parameters (Deng et al., 2026), which matches our census.

Repairing routing with data. A second line of work treats routing drift as something to fit. VS-RAQ adds value- and structure-alignment losses at 4-bit (Park et al., 2026). Hyeon and Do (2026) argue that retraining-free compression is not enough and that the router needs calibration. Both papers are the opposing regime: they spend data on the router. We spend none.

Whether protection is sufficient. The closest work to ours is *Beyond Freezing the Router* by Fang and Huang (2026), who pair rank-aligned quantization with two calibration-fitted alignment losses. They argue that freezing the router is insufficient: quantized expert outputs shift the hidden states that reach the next layer’s router, so a protected router still misroutes. Their controlled router-precision table on OLMoE appears to contradict us directly, with an FP16 router slightly *worse* than a quantized one under W4A4 experts. Their Table 8 splits that comparison by whether their alignment losses are active, and on our reading the two branches disagree. Without the losses, the FP16 router wins on OLMoE and DeepSeek-MoE and ties on Qwen3-MoE; router quantization overtakes protection only once a calibration-fitted objective has realigned it. The disagreement is therefore conditional on calibration rather than absolute. We occupy the calibration-free branch, where their own numbers favour protection, and we score it on routing behaviour, which they report only through end-task metrics, not as KL, flip rate, or load collapse.

Bit-width sweeps over MoE sub-structures. QuantMoE-Bench sweeps precision across attention, experts, block depth and linear types *inside* experts (Li et al., 2024). The router is not a condition. Its “gate projection” analysis is the FFN gate inside each expert, not the routing gate. MoQE found that expert FFNs tolerate much lower precision than attention (Kim et al., 2023). That fact motivates a follow-up in Section 5, not the main design.

4 Method

Three policies, one variable. GOLD is BF16 everywhere. UNIFORM quantizes every linear module at INT- N , including every router. MIXED uses the same INT- N policy and skips the router. Qwen’s shared_expert_gate is left quantized, so MIXED means the same thing on both models. Offline inspection of the stored router weights confirms the single-variable claim: every MIXED router is bit-identical to GOLD in every layer, and every UNIFORM router sits at the INT- N level ceiling at INT4 and INT3, and well below it at INT8.

A parameter-count-matched control. PLACEBO is UNIFORM except that one randomly chosen non-router module, matched to the total router parameter budget, stays in BF16 (placebo_seed=0). On OLMoE that module is the layer-14 expert-10 down-projection ($1.000\times$ the budget); on Qwen, the layer-19 expert-45 down-projection ($0.978\times$). The control matches parameter count and nothing else. The routers are 16 (24) small modules read by every token in every layer; the placebo is one expert down-projection in one layer, on the compute path for 11.4% (5.0%) of routing tokens. Counting token $\times$ layer events, the routers are exposed about $140\times$ and $477\times$ more. The control rules out “any high-precision parameters help.”

Four-cell attribution of the residual. Even a BF16 router can misroute if the hidden state arriving at it has already drifted through quantized attention and experts. Table 1 crosses gold versus quantized router weights with gold versus quantized incoming activations. The four logit sets are offline matrix multiplications on activations captured during the main runs: X stacks the per-token hidden states x of Equation 1 into one matrix, W is the router weight of that layer, and a cell is logits = $XW^\top$. Cell C is MIXED: the same attention and expert weights as UNIFORM, with no feedback from perturbed routing. Cell D pairs quantized router weights with those same MIXED activations, so it is **not** UNIFORM. In a real UNIFORM run a rounded router sends a token to different experts, those experts write different values into the residual stream, and every later router therefore sees a different input. Cell D holds its inputs fixed, so it measures the rounded router weights alone and omits that feedback. We score every cell against the rebuilt reference A, and we quote

	gold W	quantized W
gold X	A: reference	B: weight error
quantized X	C: activation drift (=MIXED)	D: both, no feedback ($\neq$ UNIFORM)

Table 1: The four-cell design. All four cells are logits = $XW^\top$ on captured activations. Metrics are scored against cell A.

only one share: among tokens flipped by exactly one mechanism, the fraction flipped by the router’s own weights, $B/(B+C)$. The two mechanisms are not additive (Section 6), so we do not partition D.

5 Experiments

Setup. We evaluate OLMoE-1B-7B-0924 (Muennighoff et al., 2024) and Qwen1.5-MoE-A2.7B (Qwen Team, 2024; Bai et al., 2023) on the WikiText-2 test split (Merity et al., 2017): 288,730 / 299,078 tokens, of which 288,448 / 298,785 are scored under a 1024-token window with stride 1024. Routing metrics use a seeded subset of 128 sequences $\times$ 256 tokens (32,768 tokens per layer), with 95% percentile bootstrap intervals from 200 sequence-level resamples (seed=42). We sweep INT8, INT4 and INT3. There is no calibration set. The sweep is WikiText-2 throughout; an INT4 check on C4 is reported below. Dataset construction and text examples are in Appendix C; code, configs and every run artifact behind the tables are in Appendix D.

Router protection helps, on both architectures.

Table 2 is the main result. MIXED beats UNIFORM on routing KL and top-1 flip rate at every bit-width on both models. The 95% confidence intervals never overlap (Tables 3 and 4). The effect grows as precision drops: on OLMoE, routing KL falls from 0.0085 to 0.0064 at INT8, 0.0286 to 0.0090 at INT4, and 0.1049 to 0.0385 at INT3. Qwen repeats the pattern. Perplexity moves in the same direction at INT8 and INT4. Qwen INT3 is saturation, perplexities of 9555 and 1236 against a GOLD of 7.97, and we do not treat those ratios as a quality result. Routing and collapse metrics at INT3 stay in range and remain monotone.

The effect is router-specific. PLACEBO reproduces UNIFORM on every reported metric, at all three bit-widths, on both models (Table 2). Protecting a random 0.03% of parameters is not what helps.

Follow-up: keeping attention in BF16. Figure 1 includes a later series, ATTENTION, that leaves every attention projection in BF16 and keeps the routers quantized (Kim et al., 2023). Attention is 3.88% of OLMoE and 2.81% of Qwen, about $128\times$ the router budget, so a better perplexity number is the expected tradeoff, not a rival result. Measured as the share of MIXED’s routing-KL reduction that ATTENTION recovers, $(\text{UNIFORM} - \text{ATTENTION})/(\text{UNIFORM} - \text{MIXED})$, it recovers only part of that gain at INT4 and INT3, and more than all of it at INT8 (Table 9), which is the attribution ranking of Section 6 appearing as a policy. Specificity is PLACEBO’s job, not ATTENTION’s. We expect leaving both the routers *and* attention in BF16 to improve further still, but that is a different tradeoff, and we leave it to a paper whose purpose is that mix.

Expert collapse, resolved per layer. Pooled usage entropy sits near 0.99 in every run and hides the only real collapse in the sweep. Counted per layer, a slot is starved when that expert takes under a tenth of an even share. Qwen INT3 under UNIFORM starves 222 of 1,440 slots and leaves a fully unused expert in five layers; MIXED starves 55 and leaves none unused (Figure 2, Table 7).

On OLMoE nothing collapses. The unquantized model already has 17 specialists below the line; rounding flattens load and lifts them over it, so the starved count falls as routing degrades. That is not a benefit. MIXED stays nearer GOLD than UNIFORM does. These are single-seed counts with no interval, so we rest on the Qwen gap, a factor of four, and not on a few OLMoE slots.

Where the sweep stops being informative.

Qwen INT3 is destroyed under every quantized policy (the share of positions whose most probable next token matches GOLD’s falls to 6.6% / 12.8%). The ranking among ruined models is not a quality claim. We use INT3 for routing and collapse, which stay finite and ordered, and we make perplexity claims at INT8 and INT4.

A second corpus reproduces the INT4 result.

To test whether the routing effect is a property of WikiText-2, we re-ran the INT4 cell on both models on C4 (Raffel et al., 2020), holding seed, routing-subset shape and gates fixed. The ordering repeats, including the attribution share: OLMoE routing KL 0.0273 $\rightarrow$ 0.0083, $B/(B+C)$ 63.0% (WikiText-2 62.2%); Qwen 0.0737 $\rightarrow$ 0.0272,

Model	Policy	Bits	PPL	Routing KL	Top-1 flip
OLMoE	GOLD	BF16	8.36	0.0000	0.00%
	UNIFORM / PLACEBO	INT8	9.39 / 9.39	0.0085 / 0.0085	18.01% / 18.01%
	MIXED	INT8	9.32	0.0064	13.76%
	UNIFORM / PLACEBO	INT4	10.49 / 10.49	0.0286 / 0.0286	33.01% / 33.01%
	MIXED	INT4	9.69	0.0090	16.86%
	UNIFORM / PLACEBO	INT3	30.23 / 30.23	0.1049 / 0.1049	52.52% / 52.52%
	MIXED	INT3	18.64	0.0385	32.15%
Qwen	GOLD	BF16	7.97	0.0000	0.00%
	UNIFORM / PLACEBO	INT8	9.35 / 9.35	0.0235 / 0.0235	15.65% / 15.65%
	MIXED	INT8	9.28	0.0175	12.75%
	UNIFORM / PLACEBO	INT4	12.01 / 12.01	0.0861 / 0.0861	30.62% / 30.62%
	MIXED	INT4	11.24	0.0355	17.90%
	UNIFORM / PLACEBO	INT3	9555 / 9563	0.4901 / 0.4901	72.48% / 72.48%
	MIXED	INT3	1236	0.3147	55.69%

Table 2: Main results. GOLD is the BF16 upper bound. PLACEBO is printed beside UNIFORM because the two agree to several decimals; that is the control passing. Top- k Jaccard distance tracks the flip rate in every cell (Table 8). Qwen INT3 is saturated; perplexity claims in the text use INT8 and INT4 only.

Model	Bits	UNIFORM	MIXED
OLMoE	INT8	18.01 [17.81,18.18]	13.76 [13.61,13.90]
OLMoE	INT4	33.01 [32.71,33.31]	16.86 [16.71,16.99]
OLMoE	INT3	52.52 [52.21,52.83]	32.15 [31.86,32.39]
Qwen	INT8	15.65 [15.47,15.80]	12.75 [12.58,12.90]
Qwen	INT4	30.62 [30.44,30.82]	17.90 [17.69,18.13]
Qwen	INT3	72.48 [72.11,72.85]	55.69 [55.41,55.99]

Table 3: Top-1 flip rate (%). 95% percentile intervals from 200 sequence-level resamples. Every UNIFORM/MIXED pair is disjoint.

Model	Bits	UNIFORM	MIXED
OLMoE	INT8	8.54 [8.33,8.76]	6.36 [6.16,6.55]
OLMoE	INT4	28.58 [28.10,29.10]	9.01 [8.73,9.30]
OLMoE	INT3	104.9 [102.4,107.5]	38.46 [36.77,40.16]
Qwen	INT8	23.47 [22.86,24.04]	17.48 [16.89,18.04]
Qwen	INT4	86.12 [84.86,87.38]	35.49 [34.59,36.46]
Qwen	INT3	490.1 [484.6,496.5]	314.7 [310.6,319.0]

Table 4: Routing KL ($\times 10^{-3}$), same intervals. Reductions are 25.5%, 68.5%, 63.3% on OLMoE and 25.5%, 58.8%, 35.8% on Qwen.

60.2% (58.0%). INT8 and INT3 were not repeated.

6 Attribution Analysis

End-task metrics hide how routing fails. We isolate the cause with the four-cell rebuild of Table 1: gold versus quantized router weights crossed with gold versus quantized incoming activations, as offline matrix multiplications on hidden states cached during the forward pass. That split separates the error in the router’s own weights from upstream activation drift.

The rebuild reproduces the runs. Cell A, rebuilt from stored activations and weights, matches the routing decisions the GOLD run recorded at 99.66% mean top-1 agreement on OLMoE (worst layer 99.46%) and 99.37% on Qwen (worst layer 98.49%). That floor is reconstruction noise: fp16-stored activations, recomputed in CPU fp32, against a BF16 GPU forward pass. Every difference in Table 5 is well above it. Cell C also matches the MIXED flip rates measured in the main runs of Section 5 (OLMoE INT4: 16.62% rebuilt

vs. 16.86% recorded). On top-1 flips, cell D falls a little short of UNIFORM, as it must: OLMoE INT3 is 50.01% versus 52.52%; Qwen INT3 is 68.58% versus 72.48%. The gap is the routing-feedback term that D omits. Routing KL orders the same way at INT4 and INT3, but not at INT8, where D slightly exceeds UNIFORM on both models (0.0091 versus 0.0085 on OLMoE, 0.0260 versus 0.0235 on Qwen). INT8 is where the measured quantity is smallest and therefore closest to the rebuild’s reconstruction floor, so we read that inversion as reconstruction noise rather than as a feedback term that has changed sign.

The dominant mechanism reverses between INT8 and INT4. Of the flips caused by exactly one mechanism, the router’s own weights account for 43.4% / 40.3% at INT8 (OLMoE / Qwen) and 62.2% / 58.0% at INT4 (Table 5). Above half, router protection removes the larger source. That crossover is the same on both architectures, and it is the same on routing KL: $B < C$ at INT8 and $B > C$ at INT4. The INT8 ordering is not the reconstruction noise we discounted above: that floor

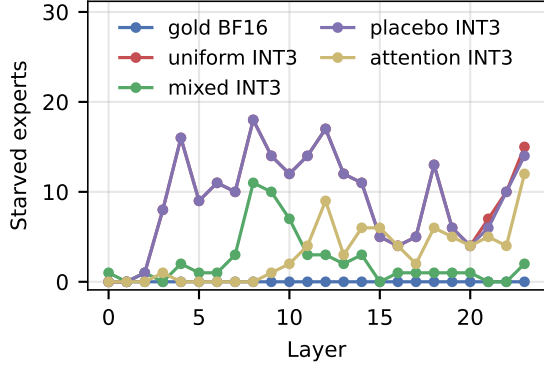


Figure 2: Starved experts by layer on Qwen at INT3, the one cell where collapse occurs. A slot is starved when an expert receives under a tenth of its even share in its own layer. UNIFORM and PLACEBO coincide, so the UNIFORM line is hidden beneath the PLACEBO one. MIXED keeps every expert alive, as does the later ATTENTION series. Damage concentrates in the mid-to-late layers, which pooled entropy averaged away. The OLMoE counterpart is flat at every depth and bit-width.

is 0.3 to 0.6% of tokens on average and at most 1.5% in any layer, while B and C differ by 3.1 and 4.2 percentage points at INT8, and the intervals on the share in Table 5 span about one point. At INT8 the cheap safeguard is fighting the smaller fire. At INT4 it is fighting the larger one. We do not read Qwen INT3 (45.4%) as activation dominance returning: that cell is the saturated regime of Section 5.

What the share does not say. The interaction residual $1 - B/D - C/D$ is large and negative everywhere (−36% to −51%). The two mechanisms flip overlapping token sets: a token that either source alone would have flipped is counted once in D and twice across B and C. B/D and C/D are therefore not shares of anything that sums to 100%, and we do not write an “ $X\% / (100 - X)\%$ ” split.

7 Discussion

Necessary but not sufficient. Leaving the routers in BF16 is a free, specific way to keep routing close to the unquantized model. It is free because the routers are 0.02% to 0.03% of the weights. It is specific because a parameter-count-matched placebo does nothing. It is not sufficient for end-task quality, because a large share of the remaining routing error, and most of the perplexity error, comes from the residual stream, not from the router matrix. A later run that also protects atten-

Model	Bits	B	C	D	$B/(B+C)$
OLMoE	INT8	10.4	13.5	17.5	43.4 [42.7,44.1]
OLMoE	INT4	27.4	16.6	31.4	62.2 [61.7,62.8]
OLMoE	INT3	43.8	31.7	50.0	58.0 [57.6,58.4]
Qwen	INT8	8.8	13.0	15.7	40.3 [39.8,40.9]
Qwen	INT4	25.0	18.1	30.3	58.0 [57.5,58.5]
Qwen	INT3*	45.8	55.1	68.6	45.4 [45.1,45.6]

Table 5: Attribution of top-1 flips (%), 10,000 bootstrap replicates. B, C and D follow Table 1. $B/(B+C)$ is the share of single-mechanism flips due to the router’s own weights. The residual $1 - B/D - C/D$ is −36% to −51% in every row, so the shares of D do not partition the drift. *Saturated.

tion improves perplexity, as expected from its size; that is extra precision on top of the router question, not an alternative answer to it.

Adjudicating the disagreement. The literature looks inconsistent only if calibration is ignored. EAQuant’s ablation shows that a more precise router helps, on a calibrated W+A baseline, scored on perplexity (Fu et al., 2025). Fang and Huang’s headline says freezing the router is not enough, and their own no-alignment split says the opposite in the calibration-free case (Fang and Huang, 2026). We measure that case on routing. Protecting the router is the right default when no data will be spent on it. It does not replace a method that realigns routing with calibration (Park et al., 2026; Hyeon and Do, 2026; Fang and Huang, 2026).

8 Limitations

The reported magnitudes are in-domain on WikiText-2; the C4 check is INT4 only. Perplexity is a point estimate and carries no interval, and the expert collapse counts of Section 5 are single-seed draws with no interval either. PLACEBO is a single draw, matched on parameter count rather than on activation exposure, and every run uses one seed. “Cheap” is a claim about parameter share and nothing else. We do not report memory or speed: these GPUs unpack the integer weights back to floats before multiplying, so a savings number would not be a real integer-kernel result, and a BF16 island inside an otherwise INT- N graph can force dtype conversions that a parameter count does not capture. That was not the question we set out to answer. DeepSeek-style routers, whose gate is a bare parameter rather than an nn.Linear (Dai et al., 2024), perform differently, and we did not evaluate their behaviour.

9 Conclusion

We asked whether, under calibration-free weight-only quantization, leaving only the MoE routers in BF16 reduces routing drift and perplexity loss. It reduces routing drift, cheaply and specifically, on two architectures and three bit-widths. The residual error switches source between INT8 and INT4, so a full-precision receptionist still misroutes if the chart on her desk is already wrong. Spending more precision on attention as well is a separate, larger tradeoff, in addition, not instead.

AI Disclosure and Reflection

We used AI assistants as additional partners, with us as the orchestrators throughout: every step was specified, reviewed and accepted by us. Gemini supported the literature search and the brainstorming that narrowed our proposal to the calibration-free, weight-only version of the router question; we went through every cited paper ourselves. Once the design was fixed we moved to Cursor (mainly Grok 4.6 and Opus 5), working step by step through the repository, the quantization and evaluation code, the analysis scripts and the tests. We also used assistance to structure and word this report and to pressure-test which claims the tables support. Every number in every table was transcribed from the run artifacts and checked against the corresponding JSON. The research question, the experimental matrix, the decision to treat the four-cell rebuild as a mechanism analysis rather than a fallback, the controls, and the interpretation are ours. The tools assisted, they did not decide.

On reflection, the assistance was most useful where the work was mechanical and checkable, such as the policy construction and the offline analyses, which we could hold to tests we had read. It was least useful where the work was generative. Draft prose reliably reached for more confidence than the tables supported, and several claims in earlier versions of this report had to be narrowed or cut once we compared them against the run artifacts, including a novelty claim that a closer reading of EAQuant’s ablation table did not survive. The habit that made the difference was treating every generated sentence about a number as a hypothesis to check rather than a result to keep.

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	OLMoE	Qwen
Layers / experts / k	16 / 64 / 8	24 / 60 / 4
Total params	6.92B	14.32B
Router params	2.10M	2.95M
Router share	0.0303%	0.0206%

Table 6: Architectures. Checkpoints: OLMoE-1B-7B-0924 (Muennighoff et al., 2024) and Qwen1.5-MoE-A2.7B (Qwen Team, 2024; Bai et al., 2023).

Model	Policy	Starved	Unused	KL
OLMoE	GOLD	17/1024	0	0.131
OLMoE	UNIFORM INT3	6/1024	0	0.163
OLMoE	MIXED INT3	9/1024	0	0.141
OLMoE	ATTENTION INT3	8/1024	0	0.142
Qwen	GOLD	0/1440	0	0.047
Qwen	UNIFORM INT3	222/1440	5	0.572
Qwen	MIXED INT3	55/1440	0	0.269
Qwen	PLACEBO INT3	220/1440	5	0.572
Qwen	ATTENTION INT3	74/1440	0	0.346

Table 7: Expert collapse, resolved per layer. A slot is starved when an expert receives under a tenth of its fair share in its own layer. Starved counts use all 32,768 routing tokens; load KL is rebuilt from the router inputs captured at one token in eight during the runs. OLMoE does not collapse; Qwen INT3 does, and MIXED mitigates it.

A Experimental setup

Table 6 records the two checkpoints. Both expose the router as `nn.Linear` named `mlp.gate`. We pin transformers to 4.x so experts remain individual linear modules rather than a fused 3D stack, which a walker that only swaps `nn.Linear` would silently skip.

WikiText-2 raw test, documents joined with a blank line and tokenized as one stream. Runs used NVIDIA GeForce RTX 2080 Ti cards (compute 7.5), three for OLMoE and five for Qwen, with torch 2.9.1, transformers 4.57.6 and torchao 0.17.0 (Or et al., 2025; Wolf et al., 2020).

B Additional policy comparisons

Two policy-level views that the main text refers to but does not have room to print. Table 8 gives the top- k Jaccard distance behind the flip rates in Table 2, and Table 9 gives the routing KL of the ATTENTION follow-up beside UNIFORM and MIXED, which is the comparison the recovery percentages in Section 5 are computed from.

Model	Bits	UNIFORM	MIXED	ATTENTION
OLMoE	INT8	0.145	0.128	0.105
OLMoE	INT4	0.236	0.149	0.206
OLMoE	INT3	0.434	0.277	0.363
Qwen	INT8	0.212	0.185	0.170
Qwen	INT4	0.356	0.244	0.321
Qwen	INT3	0.744	0.617	0.635

Table 8: Top- k Jaccard distance from the GOLD selected-expert set. GOLD is 0 by construction. The ordering matches the top-1 flip rate. The ATTENTION column is the follow-up series from Figure 1.

Model	Bits	UNIFORM	MIXED	ATTENTION	Rec.
OLMoE	INT8	0.0085	0.0064	0.0054	144%
OLMoE	INT4	0.0286	0.0090	0.0237	25%
OLMoE	INT3	0.1049	0.0385	0.0777	41%
Qwen	INT8	0.0235	0.0175	0.0165	117%
Qwen	INT4	0.0861	0.0355	0.0691	34%
Qwen	INT3*	0.4901	0.3147	0.3380	87%

Table 9: Routing KL for the ATTENTION follow-up beside the two main policies. **Rec.** is the fraction of MIXED’s reduction that ATTENTION recovers, $(\text{UNIFORM} - \text{ATTENTION}) / (\text{UNIFORM} - \text{MIXED})$, and is the quantity quoted in Section 5. Values above 100% mean ATTENTION reached a lower routing KL than MIXED, which happens only at INT8, where Table 5 puts the larger share of single-mechanism flips on the incoming activations rather than on the router’s own weights. *Saturated.

C Dataset statistics and examples

We evaluate on two language-modeling corpora.

WikiText-2. Verified Good and Featured Wikipedia articles, with original casing, punctuation and a large vocabulary (Merity et al., 2017). The test split contains 288,730 tokens on OLMoE and 299,078 on Qwen. A typical opening:

```
= Robert Boulter =
Robert Boulter is an English film ,
television and theatre actor .
He had a guest @-@ starring role on the
television series The Bill in 2000 .
```

C4. The Colossal Clean Crawled Corpus is a heuristically filtered Common Crawl scrape: ungrammatical text, boilerplate, machine-generated code and near-duplicates are removed (Raffel et al., 2020). The INT4 check uses the first 1,000 documents of the English validation shard c4-validation.00000-of-00008: 481,691 / 469,908 tokens under the OLMoE / Qwen tokenizers, of which 481,220 / 469,449 are scored, with the same 128×256 routing subset as on WikiText-2. The domain is general web text, so the routing

effect is not an artifact of Wikipedia formatting. The eleventh document of the shard opens:

```
The sport of Bocce Ball is alive and well
in Holliston.
"Throwing balls toward a target is the
oldest game known to mankind. As early
as 5000 B.C. the Egyptians played a form
of Bocce with polished rocks.
```

D Code

Code, configs, and run artifacts are at https://github.com/yoavgeva63/MoE_quantization_NLP.